

Thermal Physics — Jargon and Formula Recall Pack

Edexcel International A Level Physics · Unit 5

Prepared for Phuc Nguyen · Source-locked to audited lessons and IAL augmentation

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One mental model: temperature differences drive energy transfer from hotter to cooler regions. At particle level, internal energy, collisions and intermolecular separation explain heating, state changes and gas pressure; at macroscopic level, the correct equation, SI units, fixed conditions and graph shape turn that model into exam marks.

Source lock: the audited Alex lesson guide is primary. Edexcel Year 2 Chapters 10–12, the local SaveMyExams thermal/ideal-gas packs and IAL specification points 125–132 supply examiner wording, practical detail and full IAL gas-law context. `Phuc trap` appears only where the lesson evidence directly records Phuc’s response; all other warnings are `Common trap`.

1. Jargon glossary — foundations, radiation and transfer mechanisms

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Energy store	A place or form in which energy is accounted for before and after a transfer. The thermal store is the one central to this chapter.	A bookkeeping category for energy possessed by a system: kinetic, gravitational potential, elastic, chemical, thermal/internal, nuclear or electromagnetic. Energy is conserved between stores.	Energy E or change ΔE , joule (J).	Not a route by which energy travels.	Alex separated thermal on the stores list from heating on the pathways list.	Common trap — mixing stores and transfer pathways in one list.
Energy-transfer pathway	The way energy moves from one store or system to another.	Energy is transferred mechanically, electrically, by heating, or by radiation/waves; the total is conserved.	Energy in J; rate of transfer is power in W.	Not an energy store contained by an object.	Heating transferred energy into the aluminium block; electrical work supplied that energy.	Common trap — saying an object “stores heating.”
Internal energy	The microscopic energy inside a body because its particles move randomly and interact through bonds.	The sum of the random kinetic energies and intermolecular/interatomic potential energies of all particles in the system. For an ideal gas, intermolecular potential energy is neglected, so internal energy is kinetic.	Usually U or ΔU , J.	Not the kinetic energy of the whole object moving through a room; not a substance called “heat.”	A stationary hot block has large internal energy even though its centre of mass is not moving.	Common trap — defining internal energy as kinetic energy only for every state of matter.
Heating	Energy moving because one body or region is hotter than another.	Random energy transfer from higher temperature to lower temperature through conduction, convection or radiation, requiring a temperature difference.	Energy transferred ΔE in J; rate in W.	Not energy stored inside a body and not automatically the same as temperature.	Energy moved from Phuc’s warm hand to a cooler metal object until their temperatures approached equality.	Common trap — saying a body “contains heat”; it contains internal energy.
Temperature	A measure of how energetic the particles’ random motion is on average.	A thermodynamic state quantity directly related to mean random molecular kinetic energy; thermodynamic temperature is measured from absolute zero.	T in K; Celsius temperature often θ in $^{\circ}\text{C}$.	Not total internal energy: a small hot sample can have less total energy than a large cooler sample.	Heating a fixed-volume gas increased particle speed and therefore pressure.	Common trap — using Celsius in gas laws or confusing temperature with energy.
Thermal equilibrium	The state in which two bodies have no net thermal-energy transfer between them; if they are in thermal contact, they have the same temperature.	Thermal equilibrium requires no net transfer of thermal energy between the bodies and equality of temperature in thermal contact. Microscopic two-way exchange may continue.	Equal temperature; net thermal transfer rate between bodies = 0 W.	Not defined merely by “infrared emitted = infrared absorbed”: that is only the radiative special case and ignores conduction, convection, work or internal power.	Both cooling-can curves eventually level toward room temperature when each can reaches the surroundings’ temperature.	Common trap — replacing the general no-net-transfer/equal-temperature definition with a radiation-only rate balance.
Absolute zero	The lowest possible thermodynamic temperature.	0 K = -273.15°C , the limiting temperature at which particles have their minimum kinetic energy; Kelvin temperatures cannot be negative.	0 K.	Not ordinary “cold space”; space is about 2.7 K and a vacuum is not a perfect heat sink.	The lesson used 0 K as the limiting case in which infrared emission would cease.	Common trap — writing 0 K = 0°C .
Kelvin scale	The absolute temperature scale used whenever a law compares temperature by ratios.	A thermodynamic scale whose zero is absolute zero and whose interval has the same size as one Celsius degree: $\Delta T/\text{K} = \Delta \theta/^{\circ}\text{C}$.	T in K, never “degrees K.”	Not a different-sized degree from Celsius and not optional in gas-law ratios.	$15^{\circ}\text{C} = 288.15\text{ K}$ in the ideal-gas example.	Common trap — adding 273.15 to a temperature change ; only absolute readings require conversion.
Infrared radiation	Invisible electromagnetic radiation just beyond the red end of visible light that can transfer energy.	Ordinary thermal infrared is a broad spectrum produced by thermal motion of charged particles, including molecular rotation/vibration and lattice vibration; it needs no material medium and crosses a vacuum.	Wavelength longer and frequency lower than visible red; transfer rate in W.	Not all invisible radiation, not alpha/beta/gamma nuclear radiation, and not explained by orbital-electron drops between discrete atomic energy levels.	Herschel’s thermometer just beyond red showed the largest temperature rise.	Phuc trap — “totally forgotten” and treated invisible light too broadly; infrared is one specific EM band just beyond red.
Emission	Giving infrared energy out from a surface.	The release of electromagnetic radiation by a body; all bodies above absolute zero emit, and hotter bodies emit at a greater rate.	Emitted power in W; intensity in W m^{-2} .	Not reflection of incoming radiation and not evidence that the emitter is getting hotter.	The matt-black can cooled faster because it emitted infrared more strongly.	Phuc trap — predicted the stronger emitter’s temperature would become higher; emitting energy cools it.

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Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Absorption	Taking incoming radiation into a material so its internal energy can increase.	Transfer of incident electromagnetic energy to the absorber rather than transmission or reflection. Good absorbers are good emitters.	Absorbed energy in J; absorbed power in W.	Not reflection, which returns radiation without taking it into the material.	Earth absorbed incoming solar radiation before re-emitting at a longer wavelength.	Common trap — calling a shiny reflector a good absorber.
Reflection	Bouncing incoming radiation away from a surface.	Redirection of incident electromagnetic radiation without its energy being absorbed by that surface; shiny/light surfaces are effective infrared reflectors.	Reflectivity is dimensionless; reflected power in W.	Not emission from energy already inside the object.	Foil behind a radiator reflected infrared back into the room.	Phuc trap — thought the shiny can should “give more”; reflection needs incoming radiation, while the hot-water can comparison tested emission.
Black/matt surface	A dark, dull surface that exchanges infrared with its surroundings very effectively.	A relatively high-emissivity surface: a good emitter and, by Kirchhoff-style reciprocity, a good absorber; a poor reflector.	Emissivity ε is dimensionless if used.	Not automatically hotter than a shiny surface at the same temperature.	The black can appeared brighter on the infrared camera and cooled more steeply.	Common trap — interpreting a brighter thermogram as proof of a higher actual temperature when emissivity differs.
White/shiny surface	A light or polished surface that reflects much of the infrared reaching it.	A relatively low-emissivity, low-absorptivity surface and good reflector of thermal radiation.	Reflectivity/emissivity dimensionless if used.	Not the best emitter simply because it looks bright in visible light.	A reflective layer facing the Sun reduces incoming solar absorption on spacecraft.	Phuc trap — linked “best reflector” with “gives off more”; reflector and emitter rankings are opposite here.
Electromagnetic radiation	Energy carried by oscillating electric and magnetic fields, including visible light and infrared.	Radiation made of photons/waves spanning radio to gamma; infrared is EM radiation and can cross a vacuum.	Frequency f in Hz; wavelength λ in m; photon energy $E = hf$ if needed elsewhere.	Not synonymous with nuclear radiation; gamma is both EM and nuclear in origin, but infrared does not come from nuclear decay.	Solar/terrestrial radiation crosses space without conduction or convection.	Phuc trap — called infrared “a weak type of nuclear radiation”; Alex corrected it to electromagnetic radiation.
Radiation balance / greenhouse effect	Earth warms until outgoing energy again balances incoming energy; greenhouse gases make that balance occur at a higher temperature.	Shorter-wavelength solar radiation is absorbed by Earth; Earth re-emits longer-wavelength infrared, which greenhouse gases absorb and re-radiate, reducing escape rate until a higher-temperature equilibrium is reached.	Rates in W or W m^{-2} .	Not a solid “blanket” that creates energy and not nuclear radiation.	The board chain was Sun radiation → Earth absorption → longer-wavelength re-emission → atmospheric trapping → warming.	Common trap — omitting the longer-wavelength re-emission step or saying CO_2 “knocks radiation out.”
Conduction	Energy passing through neighbouring particles without the material flowing as a whole.	Thermal energy transfer down a temperature gradient: solids pass lattice vibrations to neighbours; gases transfer energy through molecular collisions and random molecular motion; metals also use mobile delocalised electrons.	Transfer rate in W; no single lesson formula required.	Not bulk movement of matter and not radiation across empty space.	The rod used lattice vibrations/free electrons; Sahara air touching hot ground was heated through molecular collisions.	Common trap — using the solid-lattice phrase “passing vibrations” for gas conduction; say molecular collisions/random motion in a gas.
Lattice vibration	Atoms in a solid jiggle about fixed equilibrium positions and pass energy to neighbours.	Random vibrational motion of atoms/ions in a lattice; higher-temperature regions have larger mean vibrational kinetic energy, which couples to adjacent particles.	Microscopic energy in J.	Not atoms travelling from the hot end to the cold end.	Alex’s linked-arms analogy represented vibrations passed along the solid.	Common trap — saying the particles themselves move through the solid during conduction.
Free / delocalised electron	A mobile electron in a metal that can carry energy rapidly through the lattice.	A conduction electron not bound to one atom; it gains energy near the hot region and transfers it by collisions through the metal.	Electron charge e in C if quantified; speed in m s^{-1} .	Not the ordered electron drift of an electrical current, although the same charge carriers are involved.	Metals conducted better than non-metals because free electrons supplemented vibration transfer.	Common trap — forgetting free electrons in a “why are metals good thermal conductors?” explanation.
Thermal conductor / conductivity	A material that transfers internal energy quickly when a temperature difference exists.	High thermal conduction rate for a given geometry and temperature difference; metals are usually good conductors because of delocalised electrons.	Thermal conductivity would be $\text{W m}^{-1} \text{K}^{-1}$, though no conductivity equation was required in the lesson.	Not necessarily a material at a higher temperature.	Both ice-demo blocks were about 21°C ; block 2 melted ice faster because it conducted energy faster.	Phuc trap — said the faster-melting block “must be hot”; the temperatures were the same.

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Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Thermal insulator	A material or structure that slows thermal energy transfer.	A system with low conduction and/or suppressed convection/radiation, often using few particles, trapped gas pockets, vacuum gaps or reflective surfaces. A solid's conductivity depends on lattice structure, free electrons, scattering, density and defects.	Performance may be a transfer rate in W; no single lesson formula.	Not a perfect stopper of all three mechanisms; “weaker bonds = better insulator” is not a safe universal rule.	Loft fibre trapped air; foil addressed infrared; a draught excluder blocked moving air.	Phuc trap — suggested very strong bonds make the best insulator; use the dependable sparse/no-particle logic for gas/vacuum and never rank solids from bond strength alone.
Convection	Energy carried because a liquid or gas itself moves.	Thermal transfer by bulk motion of a fluid: heating causes expansion, lower density and rise; cooler denser fluid sinks, forming circulation.	Fluid speed in m s^{-1} if measured; no core lesson formula.	Not conduction between fixed particles and not possible in a vacuum.	Heated dyed water rose and circulated around the loop.	Phuc trap — considered water boiling/becoming gas and “dropping”; the water stayed liquid and the heated region rose.
Fluid	Any substance able to flow.	A liquid or gas that can undergo bulk motion and therefore convection.	Density ρ in kg m^{-3} .	Not only a liquid; gases are fluids too.	Both sea-breeze air and dyed water formed convection currents.	Common trap — saying convection happens in gases but not liquids, or in solids.
Density	How much mass is packed into each unit volume.	Mass per unit volume, $\rho = m/V$. A heated fluid region usually expands, so the same mass occupies more volume and density falls.	ρ in kg m^{-3} .	Not pressure, and particles do not need to become physically larger.	Heated air particles spread out; the region became less dense and rose.	Phuc trap — offered mass divided by volume when asked for pressure; that is density, not pressure.
Convection current	A circulating loop in which warm, less-dense fluid rises and cooler, denser fluid sinks.	Sustained bulk circulation driven by density differences in a gravitational field, carrying internal energy.	Flow speed in m s^{-1} if quantified.	Not a change of state and not an electrical current.	Smoke descended one opening of the candle box while hot air rose over the flame.	Phuc trap — thought cooling gas must become liquid to sink; a gas can become denser while remaining gas.
Advection	Horizontal transport of warm or cool air by the wind over a large distance.	Bulk transport of a fluid property such as internal energy by the fluid's large-scale horizontal motion; unlike local buoyant convection, it does not mean the fluid rises because it became less dense.	Wind speed in m s^{-1} if quantified.	Not the local convection that first lifts heated Sahara air and not conduction between neighbouring molecules.	Local convection raised hot Sahara air; prevailing winds advected the warm air mass toward Paris.	Common trap — calling the whole Sahara-to-Paris horizontal journey “convection”; the long-range transport is advection.
Trapped air	Small pockets of air that neither conduct well nor circulate freely.	Gas with low particle number density provides poor conduction; dividing it into small immobilised pockets suppresses convection currents.	No single formula; compare transfer rates.	Not the same as a large open air gap, which can permit convection.	Loft insulation, cavity fibres and curtains trapped air; double glazing trapped air or argon.	Common trap — giving only one role; trapped air reduces both conduction and convection.
Vacuum	A region with essentially no particles.	Absence of a material medium, preventing conduction and convection; electromagnetic radiation can still pass and carry energy.	Pressure ideally approaches 0 Pa.	Not a perfect thermal insulator against radiation and not something that automatically cools a machine.	Space allowed radiative transfer from the Sun but no conduction through the intervening vacuum.	Phuc trap — suggested convection could occur if “two airs” differed; in a vacuum there is no air.
Draught	A flow of air through a gap that carries warm or cool air with it.	Bulk replacement/motion of a gas, so the relevant transfer mechanism is convection.	Air speed in m s^{-1} if measured.	Not conduction through a stationary solid.	A draught excluder under a door stopped moving air.	Phuc trap — answered “conduction”; moving/replacing air is convection.
Radiative cooling in space	Rejecting waste energy from a spacecraft by emitting infrared to empty space.	In vacuum, heat must be conducted internally to a large radiator with high thermal-infrared emissivity ; surfaces exposed to the Sun need low solar absorptivity so little incoming solar power is absorbed.	Radiated power in W; area in m^2 ; emissivity/absorptivity dimensionless.	Not convection into “cold space”; one generic “black” or “shiny” label is not enough because thermal-IR emission and solar absorption concern different wavelength bands.	Alex's design used an effective emitting radiator plus a reflective/white sunward finish.	Common trap — specify both independent requirements: high thermal-IR emissivity and low solar absorptivity.
Thermogram / infrared camera	An image built from detected infrared intensity and converted into apparent temperature colours.	A calibrated detector maps received thermal radiation; readings depend on temperature and surface emissivity.	Temperature in K or $^{\circ}\text{C}$ after calibration.	Not direct visible-light photography and not emissivity-independent proof of temperature.	The black and shiny cans at the same temperature looked different because their emission differed.	Common trap — assuming identical colour scales guarantee identical emissivity.

2. Jargon glossary — thermal calculations, state change, gases and practicals

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Specific heat capacity (SHC)	The energy needed per kilogram for each kelvin of temperature rise. A larger value means more energy for the same mass and rise.	Energy required to raise the temperature of 1 kg of a substance by 1 K without a change of state; $c = \Delta E / (m\Delta\theta)$.	c in $\text{J kg}^{-1} \text{K}^{-1}$.	Not total heat capacity of an object and not latent heat during a phase change.	Water's reference $c = 4200 \text{ J kg}^{-1} \text{K}^{-1}$ explains why it heats/cool slowly.	Common trap — using final temperature instead of temperature change, or leaving mass in grams.
“Specific” quantity	A quantity quoted per unit mass, so different sample sizes can be compared fairly.	In SHC and specific latent heat, “specific” means per kilogram.	Per kg gives factors of kg^{-1} .	Not per area or per volume.	Alex derived $\text{J}/(\text{kg K})$ directly from $\Delta E = mc\Delta\theta$.	Phuc trap — treated “specific” as possibly per area/volume; in these definitions it is per kilogram.
Heat capacity	Energy needed to raise one particular object's temperature by one kelvin.	For an object of mass m and material SHC c , heat capacity $C = mc$.	C in J K^{-1} .	Not specific heat capacity, which is per kilogram and is a material property.	A heavier aluminium block has greater heat capacity even though its c is unchanged.	Common trap — giving $\text{J kg}^{-1} \text{K}^{-1}$ as the unit of whole-object heat capacity.
Temperature change	Final temperature minus initial temperature; only the size of the interval matters in SHC.	$\Delta\theta = \theta_f - \theta_i$ or $\Delta T = T_f - T_i$; one-kelvin and one-Celsius-degree intervals are equal.	K or $^{\circ}\text{C}$ for a change.	Not the final temperature and not a ratio-scale conversion.	Cooling the avocado by 2 K gives $ \Delta\theta = 2 \text{ K}$ for the energy magnitude.	Common trap — adding 273.15 to a change or allowing a negative sign to make an energy magnitude negative.
Electrical SHC method	Supply measured electrical energy to a known mass and divide by its temperature rise.	Equate electrical work $IV\Delta t$ to the block's internal-energy increase $mc\Delta\theta$, ideally after minimising/correcting losses.	$I \text{ A}, V \text{ V}, t \text{ s}, m \text{ kg}, c \text{ J kg}^{-1} \text{K}^{-1}$.	Not exact if heater, thermometer and surroundings absorb energy.	The aluminium block used oil for good thermal contact, lagging, a heater and thermometer.	Common trap — counting all supplied energy as block energy; this usually makes measured c too large.
Thermal contact	Good physical coupling that lets energy pass efficiently between components.	Close contact that reduces interfacial thermal resistance and helps the heater/thermometer match the block temperature.	No core formula here.	Not thermal equilibrium by itself; equilibrium takes time.	Oil in the aluminium block's holes improved contact with heater and thermometer.	Common trap — leaving air gaps around sensors.
Specific latent heat	Energy per kilogram needed to change state while temperature stays constant.	Energy required to change the phase of 1 kg without temperature change; $L = \Delta E / \Delta m$.	L in J kg^{-1} .	Not energy for raising temperature within one state.	During a heating-curve plateau, supplied energy increased potential rather than kinetic energy.	Common trap — omitting “without change in temperature” from the definition.
Specific latent heat of fusion	Energy per kilogram for solid-liquid change at the melting/freezing temperature.	L_f is energy per unit mass for melting; the same magnitude is released on freezing by conservation of energy.	L_f in J kg^{-1} ; water about 3.34×10^5 .	Not vaporisation and not a temperature-rise quantity.	Ice at 0°C absorbed energy while melting but stayed at 0°C .	Common trap — applying $mc\Delta\theta$ across the melting plateau.
Specific latent heat of vaporisation	Energy per kilogram for liquid-gas change at boiling/condensing temperature.	L_v is energy per unit mass for vaporisation; the same magnitude is released in condensation.	L_v in J kg^{-1} ; water about 2.26×10^6 .	Not fusion; it is usually much larger because molecules separate much further.	For the same mass and constant power, the boiling plateau is much longer than the melting plateau because $L_v \gg L_f$.	Common trap — comparing plateau lengths without equal mass and constant-power conditions.
Change of state / phase change	A substance changes between solid, liquid and gas while supplied/removed energy changes molecular arrangement.	During a first-order phase change at fixed pressure, energy changes intermolecular potential energy while mean kinetic energy and temperature remain constant.	Energy in J; mass changing state Δm in kg.	Not necessarily a temperature change; sublimation is also a phase change though the IAL equations emphasise fusion/vaporisation.	Ice melting and water boiling formed flat sections of the heating curve.	Common trap — saying particles stop moving on a plateau.
Heating curve	A graph showing temperature as energy is supplied: slopes while solid, liquid and gas warm, with plateaus during phase changes.	Sloped sections obey $\Delta E = mc\Delta\theta$; flat sections obey $\Delta E = L\Delta m$. After boiling is complete, the gas/steam has its own rising warming section.	Vertical T or θ ; horizontal energy, or time only when power is constant.	Not one straight line and not a curve that ends at the boiling plateau.	Ice warming $\rightarrow$ melting $\rightarrow$ liquid warming $\rightarrow$ boiling $\rightarrow$ gas/steam warming.	Common trap — omit gas warming, or compare plateau lengths without the same mass and constant power.
Cooling curve	A graph of temperature falling toward surroundings, usually steep initially and flattening later.	The magnitude of gradient represents cooling rate; it decreases as the temperature difference falls. A phase change creates a near-constant-temperature region.	Gradient in K s^{-1} .	Not proof that the faster emitter becomes hotter.	The black can's curve was steeper than the silver can's but both approached room temperature.	Phuc trap — inferred strong emission should raise temperature; steeper downward gradient means faster energy loss.

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Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Pressure	Force spread over an area; gas pressure comes from molecular impacts on container walls.	Normal force per unit area, $p = F/A$. In kinetic theory, elastic wall collisions transfer momentum and create a mean force.	p in Pa, where $1 \text{ Pa} = 1 \text{ N m}^{-2}$.	Not density m/V and not merely collision count without momentum change.	Faster heated particles hit a rigid container's walls more often and harder, increasing pressure.	Phuc trap — gave mass divided by volume and said “friction into the boxes”; use force/area and wall collisions.
Fixed mass of gas	The same sealed set of particles is followed while pressure, volume or temperature changes.	Particle number N and amount n remain constant; gas-law comparisons require this condition unless the equation explicitly handles changing amount.	N molecules or n mol constant.	Not necessarily fixed volume or fixed pressure.	Boyle, Charles and pressure laws each assume a fixed mass but hold a different state variable constant.	Common trap — stating a gas law without “fixed mass.”
Ideal gas	A simplified gas whose particles are tiny, random-moving and interact only through elastic collisions.	Many point molecules in continuous random motion; negligible molecular volume and intermolecular forces except during collisions; collisions are perfectly elastic and brief. It obeys $pV = NkT$.	p Pa, V m ³ , T K.	Not every real gas under all conditions; real gases approximate it best at low density/high temperature.	The IAL gas-law and kinetic-theory equations model the fixed-volume lesson gas as ideal.	Common trap — using ideal-gas equations without Kelvin or SI volume.
Boyle's law	Compressing a fixed amount of gas at constant temperature raises pressure so pV stays constant.	For a fixed mass at constant thermodynamic temperature, $p \propto 1/V$ and $p_1 V_1 = p_2 V_2$.	p Pa and V m ³ in SI, though consistent paired units cancel in a ratio.	Not valid during uncontrolled heating and not a direct $p \propto V$ relation.	The required practical plots p against $1/V$ to obtain a straight line through the origin.	Common trap — forgetting to wait for thermal equilibrium after compression.
Isothermal	A process or curve at constant temperature.	A p - V relation for a fixed mass of gas at one fixed thermodynamic temperature; an ideal-gas isotherm is a rectangular hyperbola.	T in K, constant.	Not “insulated”; rapid compression can change temperature even if the final points later equilibrate.	Boyle's-law data were taken slowly so the gas returned to room temperature.	Common trap — treating “constant temperature” as automatically guaranteed.
Charles's law	At constant pressure, a warmer fixed mass of gas occupies a larger volume in direct proportion to Kelvin temperature.	For fixed mass and constant p , $V \propto T$ and $V_1/T_1 = V_2/T_2$.	V m ³ , T K.	Not a Celsius proportionality and not applicable in a rigid container.	A piston can move so heating at constant pressure increases gas volume.	Common trap — using °C in the ratio.
Pressure law (Gay-Lussac)	In a rigid container, heating raises pressure in direct proportion to Kelvin temperature.	For fixed mass and constant V , $p \propto T$ and $p_1/T_1 = p_2/T_2$.	p Pa, T K.	Not expansion: volume is fixed.	Alex's opening problem: faster, more frequent, harder wall collisions raise p .	Phuc trap — said the fixed-volume gas “expands”; the pressure rises instead.
Equation of state	One equation connecting the macroscopic condition of an ideal gas.	$pV = NkT = nRT$, relating pressure, volume, absolute temperature and amount of gas at equilibrium.	p Pa, V m ³ , T K, N dimensionless, n mol.	Not a transfer-rate equation and not for Celsius temperature.	The $8.3 \times 10^3 \text{ cm}^3$ cylinder example used $pV = NkT$ to find molecule number.	Common trap — mixing N molecules with n moles.
Mole	A counting unit containing Avogadro's number of particles.	Amount of substance n where one mole contains $N_A \approx 6.02 \times 10^{23}$ entities; $N = nN_A$.	n in mol.	Not the mass of one molecule or the total number N .	$pV = nRT$ and $pV = NkT$ are equivalent because $R = N_A k$.	Common trap — substituting moles for N in $pV = NkT$.
Avogadro constant	The number of particles in one mole.	$N_A \approx 6.02 \times 10^{23} \text{ mol}^{-1}$, linking microscopic particle number to macroscopic amount.	N_A in mol^{-1} .	Not Boltzmann constant and not dimensionless when written as a constant.	$N = nN_A$ converts moles to molecules.	Common trap — confusing N_A with N , the sample's actual molecule count.
Boltzmann constant	The conversion factor linking one particle's thermal energy scale to Kelvin temperature.	$k = R/N_A = 1.38 \times 10^{-23} \text{ J K}^{-1}$; it appears in $pV = NkT$ and mean molecular KE.	k in J K^{-1} .	Not the molar gas constant $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$.	The ideal-gas molecule calculation used the supplied k value.	Common trap — using R with molecule number or k with moles.
Kinetic theory	A model that explains gas behaviour through many fast, randomly moving particles and their collisions.	Connects macroscopic p , V , T to microscopic mass, momentum and mean-square speed under ideal-gas assumptions.	Uses m kg per molecule, $\langle c^2 \rangle \text{ m}^2 \text{ s}^{-2}$.	Not a claim that every molecule has the same speed or that average velocity is non-zero.	Pressure rose because heated molecules transferred momentum to walls more frequently and strongly.	Common trap — saying pressure comes from “friction” rather than momentum-changing elastic collisions.
Brownian motion	Tiny suspended particles move in random jerks because unseen molecules strike them unevenly.	Observable random motion caused by fluctuating molecular bombardment, providing evidence that matter's molecules are in continuous random motion.	Displacement in m; time in s if measured.	Not the path of a single gas molecule and not self-propelled pollen.	Textbook evidence supports the kinetic theory assumptions used in the gas derivation.	Common trap — claiming molecules themselves are visible under the microscope.

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Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Mean square speed	Square every molecular speed, then average those squared values.	$\langle c^2 \rangle = (c_1^2 + \dots + c_N^2)/N$; it enters the pressure equation.	$\text{m}^2 \text{s}^{-2}$.	Not the square of the ordinary mean speed; these are close but not identical.	Speeds $310\text{--}350 \text{ m s}^{-1}$ give $\langle c^2 \rangle = 109100 \text{ m}^2 \text{s}^{-2}$.	Common trap — averaging first and then squaring.
Root-mean-square (r.m.s.) speed	The square root of the mean of all squared molecular speeds; a useful typical random speed.	$c_{\text{rms}} = \sqrt{\langle c^2 \rangle}$; for an ideal gas $c_{\text{rms}} = \sqrt{3kT/m}$.	m s^{-1} .	Not average velocity, which is zero for a stationary gas because directions cancel.	Oxygen at 297 K with molecular mass $5.34 \times 10^{-26} \text{ kg}$ has $c_{\text{rms}} \approx 480 \text{ m s}^{-1}$.	Common trap — forgetting the square root or using molar mass where single-molecule mass is required.
Mean molecular kinetic energy	The average translational kinetic energy of one ideal-gas molecule.	$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$; it depends only on absolute temperature, not molecular identity.	J per molecule.	Not the total KE of all N molecules and not proportional to Celsius temperature.	At the same T , helium and oxygen have the same mean molecular KE but different r.m.s. speeds.	Common trap — concluding heavier molecules move faster at the same temperature.
Thermistor	A resistor whose resistance changes strongly with temperature, allowing it to be calibrated as a thermometer.	A temperature-dependent semiconductor resistance; the IAL practical records resistance against temperature while stirring and allowing equilibrium.	Resistance R in Ω ; temperature in $^\circ\text{C}$ or K .	Not a linear sensor by assumption and not self-calibrating.	Core Practical 12 plots resistance against temperature using a reference thermometer.	Common trap — heating too rapidly, not stirring, or letting measurement current self-heat the thermistor.
Control experiment in latent-heat practical	A second unheated ice funnel measures melting caused by the surroundings, so it can be subtracted.	With equal initial ice and time, corrected heater-melted mass is $\Delta m = m_A - m_B$.	Mass in kg after conversion.	Not an optional duplicate; it corrects environmental energy input.	The two-funnel method compares heated and unheated meltwater masses before $L = \Delta E / \Delta m$.	Common trap — using all melted mass as if the heater caused it.
Systematic error	A consistent bias pushing repeated results in the same direction.	An error source that shifts measurements or calculated values consistently, such as un-zeroed balance or unaccounted heat loss.	Depends on measured quantity.	Not reduced just by repeating and averaging.	Counting energy lost to surroundings as block energy makes calculated SHC systematically too large.	Common trap — calling every scatter source systematic.
Random error	Unpredictable variation causing repeated readings to scatter.	Measurement variation reduced by repeats, averaging, better resolution and controlled technique.	Often described by uncertainty in the measured unit.	Not a fixed calibration offset.	Repeating latent-heat mass measurements and averaging reduced random error.	Common trap — expecting repeats to remove a zero error.

3. Formula recall — energy, heating, pressure and core practicals

Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Pressure on an area	$p = F/A$	p pressure; F normal force; A area.	$\text{Pa} = \text{N m}^{-2}$; $\text{N}; \text{m}^2$.	Force acts normally over a known area; foundation for wall-collision pressure.	$F = pA$; $A = F/p$.	Phuc trap — m/V is density, not pressure. Convert cm^2 to m^2 .	$F = 200 \text{ N}$ over 0.040 m^2 gives $p = 5000 \text{ Pa}$.
Density	$\rho = m/V$	ρ density; m mass; V volume.	kg m^{-3} ; $\text{kg}; \text{m}^3$.	Relating mass packing to convection or kinetic pressure.	$m = \rho V$; $V = m/\rho$.	Phuc trap — do not substitute it for $p = F/A$.	0.60 kg in $5.0 \times 10^{-4} \text{ m}^3$ gives 1200 kg m^{-3} .
Fluid pressure due to depth	$\Delta p = \rho g \Delta h$	ρ fluid density; g field strength; Δh depth.	$\text{Pa}; \text{kg m}^{-3}; \text{N kg}^{-1}; \text{m}$.	Pressure increase below a fluid surface; add atmospheric pressure for total.	$\rho = \Delta p/(g \Delta h)$; $\Delta h = \Delta p/(\rho g)$.	Common trap — equation gives gauge increase, not total pressure. It may be supplied rather than memorised.	Water at 2.0 m : $\Delta p = 1000(9.81)(2) = 1.96 \times 10^4 \text{ Pa}$.
Specific heat capacity	$\Delta E = mc\Delta\theta$	ΔE energy transferred; m mass; c SHC; $\Delta\theta$ temperature change.	$\text{J}; \text{kg}; \text{J kg}^{-1} \text{K}^{-1}; \text{K}$ or $^{\circ}\text{C}$ change.	Temperature changes within one phase and no state change.	$c = \Delta E/(m\Delta\theta)$; $m = \Delta E/(c\Delta\theta)$; $\Delta\theta = \Delta E/(mc)$.	Common trap — the worksheet's printed copper $c = 3850$ is a factor-of-ten typo; accepted copper is about $385 \text{ J kg}^{-1} \text{K}^{-1}$. Preserve printed arithmetic separately, but do not learn the typo.	Printed data: $0.5 \times 3850 \times 3 = 5775 \text{ J}$. Physics-correct: $0.5 \times 385 \times 3 = 577.5 \text{ J}$.
Avocado SHC — corrected lesson value	$c = \Delta E/(m\Delta\theta)$	$\Delta E = 1294 \text{ J}$; $m = 0.215 \text{ kg}$; $\Delta\theta = 2 \text{ K}$.	$\text{J kg}^{-1} \text{K}^{-1}$.	Reconstructing board Q3 exactly.	Already rearranged for c .	Common trap — the circled 2500 does not follow from the printed numbers; re-key arithmetic and retain the raw calculator value for Q4.	$c = 1294/(0.215 \times 2) = 3009.302325 \dots \text{J kg}^{-1} \text{K}^{-1} \approx 3.01 \times 10^3 \text{ J kg}^{-1} \text{K}^{-1}$ (3 s.f., classroom convention).
Avocado temperature change — corrected continuation	$\Delta\theta = \Delta E/(mc)$	$\Delta E = 3000 \text{ J}$; $m = 0.215 \text{ kg}$; use unrounded $c = 3009.302325 \dots \text{J kg}^{-1} \text{K}^{-1}$.	K , numerically same interval in $^{\circ}\text{C}$.	Completing board Q4 using the corrected unrounded Q3 value.	From $\Delta E = mc\Delta\theta$.	Common trap — carrying the incorrect $c = 2500$ forward gives 5.58 K ; rounding Q3 early also introduces avoidable drift.	$3000/[0.215(3009.302325 \dots)] = 4.636785 \dots \text{K} \approx \mathbf{4.64 \text{ K}}$ (3 s.f.).
Heat capacity of an object	$C = mc$ and $\Delta E = C\Delta\theta$	C object heat capacity; m mass; c SHC.	C in J K^{-1} .	Comparing whole objects rather than material properties.	$m = C/c$; $c = C/m$.	Common trap — do not use $\text{J kg}^{-1} \text{K}^{-1}$ for C .	A 2.0 kg object with $c = 900$ has $C = 1800 \text{ J K}^{-1}$.
Specific latent heat	$\Delta E = L\Delta m$	L specific latent heat; Δm mass changing state.	$\text{J}; \text{J kg}^{-1}; \text{kg}$.	State change at constant temperature: fusion/freezing or vaporisation/condensation.	$L = \Delta E/\Delta m$; $\Delta m = \Delta E/L$.	Common trap — do not use $mc\Delta\theta$ on a phase-change plateau.	$10,000 \text{ J}$ melts 0.030 kg : $L = 3.33 \times 10^5 \text{ J kg}^{-1}$.
Multi-stage heating or cooling	$\Delta E_{\text{total}} = \sum mc\Delta\theta + \sum L\Delta m$	One sensible-heating term per sloped phase and one latent term per plateau.	J throughout.	A path crosses one or more phase boundaries.	Solve each stage separately, then add magnitudes.	Common trap — using one c across different phases or omitting a plateau.	Heat ice to 0°C , melt it, then heat water: $mc_{\text{ice}}\Delta\theta + mL_f + mc_{\text{water}}\Delta\theta$.
Power and energy	$P = \Delta E/\Delta t$; $\Delta E = P\Delta t$	P transfer rate; ΔE energy; Δt time.	$\text{W} = \text{J s}^{-1}$; $\text{J}; \text{s}$.	Constant-power heaters, heating-curve time intervals or radiator loads.	$\Delta t = \Delta E/P$.	Common trap — convert minutes to seconds and kW to W.	A 2.4 kW kettle for 210 s supplies $5.04 \times 10^5 \text{ J}$.
Electrical energy supplied	$\Delta E = IV\Delta t$	I current; V potential difference; Δt time.	$\text{A}; \text{V}; \text{s}$; result J .	Electrical heater energy in SHC and latent-heat practicals when I , V are measured.	$P = IV$; $I = \Delta E/(V\Delta t)$.	Common trap — symbol V here is voltage, not gas volume; not all supplied energy reaches sample.	$1.5 \text{ A} \times 10 \text{ V} \times 600 \text{ s} = 9000 \text{ J}$.

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Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
SHC from electrical block practical	$c = IV\Delta t / [m(\theta_f - \theta_i)]$	Electrical input divided by mass and measured temperature rise.	$\text{J kg}^{-1} \text{K}^{-1}$.	Aluminium-block method after measuring $I, V, t, m, \theta_i, \theta_f$.	Equivalent to $IVt = mc\Delta\theta$.	Common trap — heat losses/heater heating make the calculated c too large unless corrected; use oil/lagging.	$I = 1.5, V = 10, t = 600,$ $m = 1.0, \Delta\theta = 10$ gives $c = 900 \text{ J kg}^{-1} \text{K}^{-1}$.
Latent-heat control correction	$\Delta m = m_A - m_B$ then $L = \Delta E / \Delta m$	m_A heated-funnel melt; m_B control melt due to surroundings.	kg and J kg^{-1} .	Two-funnel Core Practical 13.	$\Delta E = L(m_A - m_B)$.	Common trap — convert grams to kg and subtract control melt before division.	$m_A = 40 \text{ g}, m_B = 10 \text{ g}$ gives $\Delta m = 0.030 \text{ kg}$; 10 kJ gives $3.33 \times 10^5 \text{ J kg}^{-1}$.

4. Formula recall — gas laws, ideal gases and kinetic theory

Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Celsius to kelvin	$T/K = \theta/^{\circ}\text{C} + 273.15$	T thermodynamic temperature; θ Celsius temperature.	K and $^{\circ}\text{C}$.	Every gas-law or molecular-KE calculation using absolute temperature.	$\theta/^{\circ}\text{C} = T/K - 273.15$.	Common trap — Kelvin has no degree sign; 15°C is not 15 K.	$15^{\circ}\text{C} = 288.15\text{ K}$; $300\text{ K} = 26.85^{\circ}\text{C}$.
Temperature interval equivalence	$\Delta T/K = \Delta \theta/^{\circ}\text{C}$	Same-sized interval on both scales.	K or $^{\circ}\text{C}$ for a difference.	SHC and any temperature change rather than absolute ratio.	None needed.	Common trap — do not add 273.15 to $\Delta\theta$.	$80 - 20 = 60^{\circ}\text{C}$ change = 60 K change.
Boyle's law — constant T	$pV = \text{constant};$ $p_1V_1 = p_2V_2$	Initial/final pressure and volume for fixed mass.	Usually Pa and m^3 ; consistent same units can cancel.	Fixed mass, constant thermodynamic temperature.	$p_2 = p_1V_1/V_2;$ $V_2 = p_1V_1/p_2$.	Common trap — condition “constant temperature” is part of the law; use total pressure where required.	$100\text{ kPa} \times 400\text{ cm}^3 = p_2(200\text{ cm}^3)$, so $p_2 = 200\text{ kPa}$.
Charles's law — constant p	$V/T = \text{constant};$ $V_1/T_1 = V_2/T_2$	Volumes and absolute temperatures for fixed mass.	V consistent; T K.	Piston/free expansion at fixed pressure.	$V_2 = V_1T_2/T_1;$ $T_2 = T_1V_2/V_1$.	Common trap — Celsius ratios are invalid.	300 cm^3 at 293 K heated to 373 K gives $V_2 \approx 382\text{ cm}^3$.
Pressure law — constant V	$p/T = \text{constant};$ $p_1/T_1 = p_2/T_2$	Pressures and absolute temperatures for fixed mass, rigid volume.	p consistent; T K.	Sealed rigid container or approximately fixed tyre volume.	$p_2 = p_1T_2/T_1;$ $T_2 = p_2T_1/p_1$.	Phuc trap — fixed volume means no expansion; pressure rises.	101 kPa at 293 K heated to 353 K gives $p_2 \approx 122\text{ kPa}$.
Combined gas law — fixed mass	$p_1V_1/T_1 = p_2V_2/T_2$	Initial/final p, V, T ; same gas amount.	T K; use consistent p, V units.	More than one of pressure, volume and temperature changes.	$V_2 = p_1V_1T_2/(p_2T_1)$, similarly for others.	Common trap — source textbook example printed a denominator typo in prose; dimensional cross-multiplication prevents it.	If $p_1 = 200\text{ kPa}$, $V_1 = 300\text{ cm}^3$, $T_1 = 300\text{ K}$, $p_2 = 100\text{ kPa}$, $T_2 = 350\text{ K}$, then $V_2 = 700\text{ cm}^3$.
Ideal gas — molecule form	$pV = NkT$	N number of molecules; k Boltzmann constant; p, V, T state variables.	Pa, m^3 , K; $k = 1.38 \times 10^{-23}\text{ J K}^{-1}$.	Particle-count ideal-gas problems.	$N = pV/(kT);$ $T = pV/(Nk)$.	Common trap — SI volume and kelvin are compulsory; N is a count, not moles.	$p = 4.5 \times 10^7\text{ Pa}$, $V = 8.3 \times 10^{-3}\text{ m}^3$, $T = 288.15\text{ K}$ gives $N = 9.4 \times 10^{25}$.
Ideal gas — mole form	$pV = nRT$	n amount in moles; R universal molar gas constant.	$R = 8.31\text{ J mol}^{-1}\text{ K}^{-1}$.	Moles are given rather than molecule count.	$n = pV/(RT);$ $T = pV/(nR)$.	Common trap — do not use R with N molecules.	1.0 mol at 273 K and 101 kPa gives about $0.0224\text{ m}^3 = 22.4\text{ L}$.
Particle-mole links	$N = nN_A; k = R/N_A$	N_A Avogadro constant.	$N_A = 6.02 \times 10^{23}\text{ mol}^{-1}$.	Converting between the two ideal-gas forms.	$n = N/N_A; R = kN_A$.	Common trap — keep symbols N, n, N_A distinct.	2.0 mol contains 1.20×10^{24} molecules.
Kinetic pressure equation	$pV = \frac{1}{3}Nm\langle c^2 \rangle$	m mass of one molecule; $\langle c^2 \rangle$ mean square speed.	J on each side; m kg; $\langle c^2 \rangle\text{ m}^2\text{ s}^{-2}$.	Deriving pressure from microscopic collisions; IAL derivation context.	$\langle c^2 \rangle = 3pV/(Nm)$.	Common trap — m is one molecule's mass, while Nm is total gas mass; factor 1/3 comes from random 3-D motion.	Equating this with NkT gives $m\langle c^2 \rangle = 3kT$.
Kinetic pressure using density	$p = \frac{1}{3}\rho\langle c^2 \rangle$	$\rho = Nm/V$ gas density.	Pa, kg m^{-3} , $\text{m}^2\text{ s}^{-2}$.	Pressure and gas density are known and r.m.s. speed is required.	$c_{\text{rms}} = \sqrt{3p/\rho}; \rho = 3p/\langle c^2 \rangle$.	Common trap — take the square root for speed.	Air: $p = 101.325\text{ kPa}$, $\rho = 1.2250$ gives $c_{\text{rms}} \approx 498\text{ m s}^{-1}$.
Mean square speed from data	$\langle c^2 \rangle = (\sum c_i^2)/N$	Individual molecular speeds c_i ; number sampled N .	$\text{m}^2\text{ s}^{-2}$.	A list/distribution of particle speeds is supplied.	$c_{\text{rms}} = \sqrt{\langle c^2 \rangle}$.	Common trap — not (mean speed) 2 .	310, 320, 330, 340, 350 m s^{-1} give $109100\text{ m}^2\text{ s}^{-2}$.
Root-mean-square speed	$c_{\text{rms}} = \sqrt{\langle c^2 \rangle} = \sqrt{3kT/m}$	m one molecule's mass; T absolute temperature.	m s^{-1} .	Molecular mass and temperature are supplied.	$T = mc_{\text{rms}}^2/(3k);$ $m = 3kT/c_{\text{rms}}^2$.	Common trap — use kg per molecule, not kg per mole; $c_{\text{rms}} \propto \sqrt{T}$.	Oxygen at 297 K, $m = 5.34 \times 10^{-26}\text{ kg}$; $c_{\text{rms}} \approx 480\text{ m s}^{-1}$.
Mean KE of one ideal-gas molecule	$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	Left is mean translational KE per molecule; right links to absolute temperature.	J per molecule.	IAL spec point 132 derivation/use and comparisons at equal T .	$T = m\langle c^2 \rangle/(3k);$ $\langle c^2 \rangle = 3kT/m$.	Common trap — must derive/use, with T in K; same T means same mean KE, not same speed.	At 300 K, mean KE = $1.5(1.38 \times 10^{-23})(300) = 6.21 \times 10^{-21}\text{ J}$.
Total translational KE of N ideal-gas molecules	$E_k = \frac{3}{2}NkT$	Multiply mean molecular KE by molecule number.	J.	Total translational internal energy for a monatomic ideal gas in this simple model.	$N = 2E_k/(3kT)$.	Common trap — molecular rotation/vibration can add internal-energy modes for non-monatomic gases.	$N = 10^{20}$ at 300 K gives $E_k \approx 0.621\text{ J}$.

5. Formula recall — SI conversions and notation

Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Mass: grams to kilograms	$m(\text{kg}) = m(\text{g}) \times 10^{-3}$	1000 g = 1 kg.	kg required in SI thermal/gas formulas.	Every $mc\Delta\theta$ or $L\Delta m$ substitution.	kg to g: $\times 10^3$.	Common trap — treating 215 g as 215 kg.	215 g = 0.215 kg.
Volume: cubic centimetres to cubic metres	$V(\text{m}^3) = V(\text{cm}^3) \times 10^{-6}$	Cubing 1 cm = 10^{-2} m gives 10^{-6} .	m^3 .	Ideal-gas equation and density in SI.	m^3 to cm^3 : $\times 10^6$.	Common trap — using 10^{-2} or 10^{-3} because “centi” is seen once.	$8.3 \times 10^3 \text{ cm}^3 = 8.3 \times 10^{-3} \text{ m}^3$.
Volume: litres to cubic metres	1 L = 10^{-3} m^3	Also 1 L = 1000 cm^3 .	L or m^3 .	Gas volumes and water-mass approximations.	m^3 to L: $\times 10^3$.	Common trap — one millilitre is one cm^3 , not one m^3 .	1.5 L = $1.5 \times 10^{-3} \text{ m}^3$.
Area: square centimetres to square metres	$A(\text{m}^2) = A(\text{cm}^2) \times 10^{-4}$	Squaring 10^{-2} gives 10^{-4} .	m^2 .	Pressure F/A .	m^2 to cm^2 : $\times 10^4$.	Common trap — using 10^{-2} .	$0.80 \text{ cm}^2 = 8.0 \times 10^{-5} \text{ m}^2$.
Pressure prefixes	1 kPa = 10^3 Pa; 1 MPa = 10^6 Pa	kilo 10^3 ; mega 10^6 .	Pa.	Ideal-gas SI substitutions and fluid pressure.	Pa to kPa: $\div 10^3$.	Common trap — gauge pressure may need atmospheric pressure added before a gas law.	101 kPa = 1.01×10^5 Pa.
Energy prefixes	1 kJ = 10^3 J; 1 MJ = 10^6 J	kilo and mega prefixes.	J.	SHC/latent values and heater energy.	J to MJ: $\div 10^6$.	Common trap — mixing a latent heat in MJ/kg with energy in J.	0.6 MJ = 6.0×10^5 J.
Power prefixes	1 kW = 10^3 W	Power is energy per second.	$\text{W} = \text{J s}^{-1}$.	Heating time from $E = Pt$.	W to kW: $\div 10^3$.	Common trap — kW is not kJ unless a time is included.	2.4 kW = 2400 W.
Time: minutes to seconds	$t(\text{s}) = 60t(\text{min})$	SI time unit is second.	s.	Electrical energy IVt , power, graph gradients.	s to min: $\div 60$.	Common trap — a gradient in K/min must be divided by 60 to obtain K/s.	12.0 min = 720 s.
SHC unit equivalence	$\text{J kg}^{-1} \text{ K}^{-1} = \text{J}/(\text{kg K})$	Energy per mass per temperature interval.	Both notations equivalent.	Definition and unit derivation.	From $c = \Delta E/(m\Delta\theta)$.	Common trap — omitting either kg or K inverse factor.	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$ means 4200 J per kg per K.
Latent-heat unit equivalence	$\text{J kg}^{-1} = \text{J}/\text{kg}$	Energy per mass changing state.	J kg^{-1} .	Definition and unit derivation.	From $L = \Delta E/\Delta m$.	Common trap — adding K^{-1} even though temperature is constant.	$L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$.
Pressure unit identity	$1 \text{ Pa} = 1 \text{ N m}^{-2} = 1 \text{ J m}^{-3}$	The last identity follows because N m is J.	Pa.	Unit checks for pV as energy.	pV has units J.	Common trap — writing N/m rather than N/m^2 .	$pV = (\text{N m}^{-2})(\text{m}^3) = \text{N m} = \text{J}$.

6. Formula recall — graph shapes and experimental recognition

Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Black vs silver cooling curves	$ d\theta/dt _{black} > d\theta/dt _{silver}$ initially	Gradient magnitude is cooling rate; same start temperature and geometry.	K s^{-1} or $^{\circ}\text{C s}^{-1}$.	Inferring relative infrared emission from temperature-time data.	Steeper negative gradient means greater energy-loss rate.	Phuc trap — stronger emission lowers temperature faster; it does not make the can hotter.	Starting near 80°C , black falls more steeply; both asymptote to room temperature.
General cooling curve	$d\theta/dt < 0$ and its magnitude decreases toward equilibrium	Temperature difference drives net loss; source guide uses qualitative exponential-like decay.	Temperature vs time.	Sketching an object cooling in fixed surroundings.	At equilibrium gradient $\rightarrow 0$.	Common trap — drawing a straight line through room temperature and continuing below without a new mechanism.	Curve is steep early, then flattens toward ambient temperature.
Heating curve: one phase	$\Delta E = mc\Delta\theta$	Sloped segment; if input power is constant, slope depends on $P/(mc)$.	Temperature vs energy or time.	Solid, liquid or gas warms without changing phase, including gas/steam warming after boiling.	$d\theta/dt = P/(mc)$ for ideal constant-power/no-loss case.	Common trap — stopping the curve at boiling; include the final rising gas-warming segment.	Solid warms, liquid warms, and after the boiling plateau steam warms on a third slope.
Heating curve: phase-change plateau	$\Delta E = L\Delta m$ and $\Delta\theta = 0$	Horizontal segment while mass changes state.	Temperature constant; horizontal energy/time.	Melting/freezing or boiling/condensing.	Plateau duration $t = L\Delta m/P$ at constant power.	Common trap — flat does not mean no energy transfer; plateau-length comparisons require the same mass and constant power.	For equal mass at constant power, water's boiling plateau is longer than melting because $L_v \gg L_f$.
Boyle graph: p against V	$p \propto 1/V$	Fixed mass, constant T .	Pa vs m^3 .	Recognising an isotherm.	Product pV constant at every point.	Common trap — not a straight decreasing line; it is a rectangular hyperbola.	Halving V doubles p .
Boyle linearisation: p against $1/V$	$p = (\text{constant})(1/V)$	Gradient is pV for that fixed mass and temperature.	Pa vs m^{-3} .	Core Practical 14 confirmation of Boyle's law.	Straight line through origin supports $p \propto 1/V$.	Common trap — plotting p against V and expecting a straight line.	Double $1/V$ gives double p .
Pressure law graph	$p \propto T$ at fixed V	T is kelvin temperature.	Pa vs K .	Gas pressure-temperature experiment.	Straight line through origin on a p - T plot.	Common trap — p against Celsius extrapolates to zero near -273.15°C , not at 0°C .	Doubling T in kelvin doubles absolute pressure.
Charles graph	$V \propto T$ at fixed p	T in K.	$\text{m}^3 \text{ vs K}$.	Constant-pressure gas expansion.	Straight line through origin ideally.	Common trap — a V -Celsius line does not pass through the origin.	From 300 K to 450 K, ideal V rises by factor 1.5.
r.m.s. speed vs temperature	$c_{r.m.s} \propto \sqrt{T}$ for fixed molecular mass	From $c_{r.m.s} = \sqrt{3kT/m}$.	$\text{m s}^{-1} \text{ vs K}$.	Comparing molecular speeds at two temperatures.	$c_2/c_1 = \sqrt{T_2/T_1}$.	Common trap — speed is not directly proportional to T .	Quadrupling Kelvin temperature doubles r.m.s. speed.
Thermistor thermostat calibration	Calibrate the potential-divider output against temperature	Thermistor and fixed resistor form a potential divider; for an NTC thermistor, resistance falls as temperature rises, changing the output voltage used to switch at a chosen set point.	Resistance in Ω ; output voltage in V; temperature in $^{\circ}\text{C}$ or K.	Core Practical 12, exact spec point 126.	Read the calibrated voltage corresponding to the desired thermostat temperature.	Common trap — assuming a universal linear R - T curve or omitting the potential-divider/thermostat purpose.	Record divider output against a reference thermometer, then choose the voltage threshold for the set temperature.

7. Formula-given vs concept-memorise map

Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Given/use — SHC	$\Delta E = mc\Delta\theta$; material c value supplied	Know every symbol and definition of c .	J, kg, $\text{J kg}^{-1} \text{K}^{-1}$, K.	IAL spec point 125 calculations.	All three standard rearrangements must be fluent.	Concept to memorise — “energy for 1 kg by 1 K,” SI mass and temperature change.	Avocado raw 3009.302325 $\text{J kg}^{-1} \text{K}^{-1}$ (3 s.f.).
Given/use — latent heat	$\Delta E = L\Delta m$	Distinguish L_f and L_v .	J, J kg^{-1} , kg.	IAL spec point 125 and phase-change stages.	$L = \Delta E / \Delta m$.	Concept to memorise — no temperature change; energy changes potential energy.	Choose L_v for boiling/condensing.
Given/use — ideal gas	$pV = NkT$	N molecules and T in K; mole form $pV = nRT$ is useful context.	SI for direct substitution.	IAL spec point 130.	Rearrange algebraically for any unknown.	Concept to memorise — ideal-gas assumptions, fixed amount, Kelvin and SI conversions.	Cylinder example gives 9.4×10^{25} molecules.
Derive and use — molecular KE	$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	Mean KE of one molecule and absolute temperature.	J and K.	IAL spec point 132.	Derive by equating $pV = \frac{1}{3}Nm\langle c^2 \rangle$ with NkT .	Concept to memorise — mean KE $\propto T$ and equal T gives equal mean KE.	Oxygen at 297 K has r.m.s. speed about 480 m s^{-1} .
Spec 125 — SHC and latent-heat equations	$\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$	Use both energy equations.	J, kg, K, $\text{J kg}^{-1} \text{K}^{-1}$, J kg^{-1} .	Exact official IAL mapping.	Rearrange each for any unknown.	Mapping trap — both equations are point 125, not points 127/128.	SHC on slopes; latent heat on plateaus.
Spec 126 — Core Practical 12	Calibrate a thermistor in a potential-divider circuit as a thermostat	Output voltage changes with thermistor resistance and temperature.	V, Ω , $^\circ\text{C}$ or K.	Exact official IAL mapping.	Use calibration to select a switching voltage.	Mapping trap — CP12 is 126.	Map divider output to the thermostat set temperature.
Spec 127 — Core Practical 13	Determine the specific latent heat of a phase change	Measure electrical energy and corrected mass changed.	J and kg, giving L in J kg^{-1} .	Exact official IAL mapping.	$L = IV\Delta t / \Delta m$ after control correction.	Mapping trap — CP13 is 127; the equation itself is point 125.	Subtract control melt before calculating L .
Spec 128 — internal energy	$U = \sum(E_{k,\text{random}} + E_p)$ conceptually	Random kinetic plus potential energies distributed among molecules.	J.	Exact official IAL mapping.	No single required rearrangement.	Mapping trap — internal energy is 128, not latent heat.	Heating one phase raises random KE; phase change raises potential energy.
Spec 129 — absolute zero and mean KE	Mean molecular kinetic energy relates to absolute temperature	$T = 0 \text{ K}$ is the minimum-kinetic-energy limit.	K and J.	Exact official IAL mapping.	Convert θ to T before ratios.	Mapping trap — absolute zero is 129.	$0 \text{ K} = -273.15 \text{ }^\circ\text{C}$.
Spec 130 — ideal gas	$pV = NkT$	N molecules; k Boltzmann constant; T kelvin.	Pa, m^3 , K, J K^{-1} .	Exact official IAL mapping.	$N = pV / (kT)$ and other rearrangements.	Mapping trap — $pV = NkT$ is 130.	Use SI volume and kelvin.
Spec 131 — Core Practical 14	Investigate p against V at fixed temperature	Boyle relation $pV = \text{constant}$; linearise with p against $1/V$.	Pa and m^3 .	Exact official IAL mapping.	$p_1V_1 = p_2V_2$.	Mapping trap — CP14 is 131.	A straight p - $1/V$ graph through the origin supports Boyle’s law.
Spec 132 — derive/use molecular KE	$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	Derive using $pV = \frac{1}{3}Nm\langle c^2 \rangle$ and $pV = NkT$.	J per molecule.	Exact official IAL mapping.	$c_{\text{rms}} = \sqrt{3kT/m}$.	Mapping trap — molecular KE derive/use is 132.	Equal T gives equal mean molecular KE.
Know/use — gas-law cases	Boyle: pV constant; Charles: V/T constant; pressure: p/T constant	Each law holds a different variable constant for fixed mass.	T always K in ratios.	Choosing a two-state gas relation.	Cross-multiply only after stating fixed conditions.	Concept to memorise — conditions are part of every law and total pressure may be needed.	Rigid heated gas uses pressure law, not Charles.
May be supplied — fluid pressure	$\Delta p = \rho g \Delta h$	Pressure increase due to fluid column.	Pa.	Synoptic pressure/gas examples.	Add atmospheric pressure for total.	Concept to memorise — gauge vs total pressure and SI dimensions.	At depth, $p_{\text{total}} = p_{\text{atm}} + \rho g \Delta h$.
Memorise — transfer mechanisms	No single formula replaces the causal chains	Solid conduction: lattice vibrations/free electrons; gas conduction: molecular collisions/random motion; convection: local buoyant fluid flow; advection: large-scale horizontal transport; radiation: EM.	Explanations use correct physical quantities and units where relevant.	Worded thermal-transfer, insulation and atmospheric-transport questions.	Not applicable.	Required recall — choose the carrier and scale precisely.	Sahara ground heats adjacent air by conduction, local convection lifts it, and wind advects it horizontally to Paris.

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Quantity/case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Memorise — surface rules	Good emitter $\leftrightarrow$ good absorber at the same wavelength; reflection opposes absorption	Emission, absorption and reflection are distinct processes.	Qualitative unless emissivity/absorptivity is introduced.	Cooling cans, thermograms, house foil and spacecraft design.	Not applicable.	Required recall — spacecraft need high thermal-IR emissivity and low solar absorptivity, properties in different wavelength bands.	Radiator rejects thermal IR efficiently while its solar-facing finish absorbs little sunlight.
Memorise — equilibrium and absolute zero	No net thermal-energy transfer between bodies; equal temperature in thermal contact; $0\text{ K} = -273.15^\circ\text{C}$	Constant temperature can coexist with microscopic two-way exchange. “IR emitted = IR absorbed” is only the radiative special case.	K and net transfer rate in W.	General equilibrium, radiation balance, cooling endpoints and gas graphs.	Not applicable.	Required recall — use the general no-net-transfer/equal-temperature definition; Kelvin cannot be negative.	Cooling curves level when can and room reach the same temperature.
Memorise — trapped-air insulation	No single equation in lesson scope	Few particles slow conduction; immobilised pockets suppress convection.	Qualitative transfer rate.	Loft, cavity wall, glazing, curtain questions.	Not applicable.	Required recall — state both effects and identify foil/radiation separately.	Draught excluder reduces convection; foil reflects infrared.
Memorise — practical error direction	$c_{calc} = E_{counted}/(m\Delta\theta)$ explains bias	If counted energy exceeds energy entering sample, c_{calc} is too large.	$\text{J kg}^{-1}\text{K}^{-1}$.	Evaluating aluminium-block SHC practical.	Correct by insulation, contact, calibration or loss correction.	Required recall — repeats do not remove systematic heat-loss bias.	Heater/surroundings take energy, so block rise is too small for counted input.
Memorise — graph recognition	Slopes use $mc\Delta\theta$; plateaus use $L\Delta m$; p - V hyperbola; p - $1/V$ and p - T straight	Match axes, fixed conditions and gradient meanings.	Axis-dependent.	Sketching and interpreting thermal/gas graphs.	State proportionality before shape.	Required recall — Celsius gas plots generally do not pass through origin.	p - T in K is straight through origin at fixed volume.

8. Final mechanism and calculation checklist

- **Worded mechanism:** name the mechanism, state the microscopic/bulk carrier, give the causal chain in the correct order, and finish with energy moving hot $\rightarrow$ cold.
- **Real scenario:** do not force one mechanism when several operate. The Sahara ground received radiation, heated adjacent air by molecular-collision conduction, local convection lifted it, and prevailing winds advected the warm air horizontally to Paris.
- **Calculation layout:** write the target symbol, write the equation, convert to SI, substitute with units, calculate, round sensibly, and check physical direction.
- **State change:** split the path into sloped $mc\Delta\theta$ stages and flat $L\Delta m$ stages; add energies.
- **Gas law:** write **fixed mass**, identify the constant variable, convert every absolute temperature to kelvin, and distinguish gauge from total pressure.
- **Space:** vacuum removes conduction and convection, not radiation; specify a radiator with high thermal-IR emissivity and a solar-facing finish with low solar absorptivity.
- **Evidence labels:** avocado Q3 is raw 3009.302325... $\text{J kg}^{-1} \text{K}^{-1}$, reported as $3.01 \times 10^3 \text{J kg}^{-1} \text{K}^{-1}$ (3 s.f., classroom convention); Q4 is $\Delta T \approx 4.64 \text{K}$.